

Microeconomics

Lecture 6: Technology and the Production Function

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What we know so far

- Consumer's optimal choice: preferences + constraint $\Rightarrow$ demand
- Comparative statics: how the optimum changes when prices/income change
- Elasticity: a % measure of responsiveness (useful for business decisions)

Next step

We now turn to **producer theory**: input $\rightarrow$ output. **Technology** is the constraint (like the budget for the consumer).

Learning objectives

After this lecture you will be able to:

1. Define the **production set** and the **production function** $Q = f(L, K)$
2. Compute and interpret the **marginal product** of labor MPL and of capital MPK
3. Understand **diminishing marginal returns** in the short run (when an input is fixed)
4. Connect these concepts to real business choices (capacity, bottlenecks, investment)
5. Solve basic numerical exercises on MPL/MPK and discrete changes

Focus of the lecture

Understand **why** “hiring more and more people” does not raise output proportionally if capacity does not change.

Technology: definition

Technology describes **the physical/organizational constraints** of the input $\rightarrow$ output transformation.

Production function

$$Q = f(L, K)$$

- L = labor (hours, workers, teams)
- K = capital (machines, plants, software, capacity)
- Q = output (units produced, services delivered)

Key idea

As for the consumer: **you cannot choose just any point**, you are constrained (here by technology).

Production set (intuition)

Production set

The set of all **feasible** combinations (L, K, Q) .

Example (services): a call center has L operators and K workstations/lines.

- if K is low, even many operators do not raise the number of calls handled by much;
- investing in K can “unlock” labor productivity.

Managerial message

Low output does not mean “poor workers”: it can be a **capacity bottleneck**.

Numerical example: production table (discrete)

Output Q (units/hour) as L and K vary (illustrative numbers):

	$K = 1$	$K = 2$	$K = 3$
$L = 1$	50	80	100
$L = 2$	80	120	150
$L = 3$	95	140	175
$L = 4$	105	150	185

Question

If I increase L by 1 with K fixed, by how much does Q increase?

Marginal product of labor (MPL)

$$MPL(L, K) = \frac{\partial f(L, K)}{\partial L}$$

Interpretation

Additional output generated by **1 more unit of labor**, holding K fixed.

In practice (discrete table):

$$MPL \approx f(L+1, K) - f(L, K)$$

Marginal product of capital (MPK)

$$MPK(L, K) = \frac{\partial f(L, K)}{\partial K}$$

Interpretation

Additional output generated by **1 more unit of capital**, holding L fixed.

In practice:

$$MPK \approx f(L, K+1) - f(L, K)$$

Exercise

Use the previous table.

1. Compute MPL with $K = 1$: from $L = 1 \rightarrow 2$, $2 \rightarrow 3$, $3 \rightarrow 4$.
2. Compute MPK with $L = 3$: from $K = 1 \rightarrow 2$, $2 \rightarrow 3$.
3. Which input is “scarcer” (and should therefore be increased) at the point $(L, K) = (3, 1)$?

Solution

With $K = 1$:

$$MPL_{1 \rightarrow 2} = 80 - 50 = 30$$

$$MPL_{2 \rightarrow 3} = 95 - 80 = 15$$

$$MPL_{3 \rightarrow 4} = 105 - 95 = 10$$

With $L = 3$:

$$MPK_{1 \rightarrow 2} = 140 - 95 = 45$$

$$MPK_{2 \rightarrow 3} = 175 - 140 = 35$$

Interpretation

At $(3, 1)$ capital is relatively scarcer: adding it raises output a lot (high MPK).

Short run: at least one fixed input

Short run

At least one input is **fixed** (typically K).

With K fixed, we typically observe:

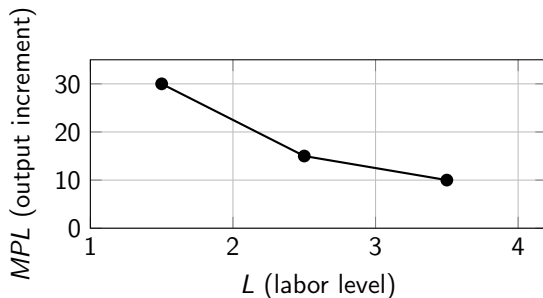
$MPL(L, K)$ decreases as L increases

- congestion/coordination costs
- bottlenecks (capacity)
- imperfect organization

Graph: diminishing *MPL* (from the table)

We derive *MPL* for $K = 1$ from the table:

MPL : 30, 15, 10 (for $L : 1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4$)



Intuition

At first you “fill” unused capacity: then capacity becomes binding and marginal productivity falls.

Two concrete examples

1) Assembly line

- K = number of workstations/machines
- L = workers on the line

If you increase L beyond the number of workstations, workers get in each other's way: MPL falls.

2) Restaurant kitchen

- K = stoves/ovens/space
- L = cooks/kitchen assistants

In a small kitchen, one more cook helps little:
bottleneck = space/equipment.

Message

If you want to raise Q a lot, **hiring is often not enough**: you must increase capacity (K).

Why do we also talk about TP , AP , MP ?

To properly understand the cost curves in the next lectures.

- **TP (Total Product)**: Q total product
- **AP (Average Product)**: average product of labor $AP_L = \frac{Q}{L}$
- **MP (Marginal Product)**: marginal product of labor MP_L

Key link

When MP_L falls a lot, producing “one more unit” requires a lot of labor $\Rightarrow$ **marginal cost rises**.

Example: TP , AP , MP table (K fixed)

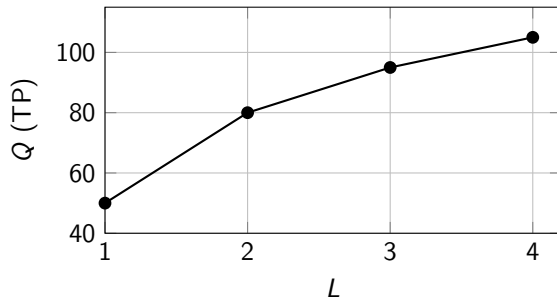
Suppose K is fixed and we observe Q as L varies:

L	1	2	3	4
Q (TP)	50	80	95	105
$AP = Q/L$	50	40	31.7	26.25
MP	–	30	15	10

Question

How do AP and MP move as L increases?

Schematic graph: TP rises but at a decreasing rate



Intuition

Q always increases, but the increment (i.e., MP) decreases: “concave” growth.

Useful rule: relationship between MP and AP

Rule

- If $MP > AP$ then AP **rises**
- If $MP < AP$ then AP **falls**
- If $MP = AP$ then AP is at its **maximum**

Analogy (very useful)

It is the same logic as “**next grade** vs **average**”.

Exercises

What we will cover:

1. MPL from a table (marginal returns)
2. MPK from a table (bottleneck)
3. Cobb–Douglas: MPL , MPK and interpretation
4. Managerial decision: hire vs invest

Exercise—MPL from a table (5 min)

With $K = 1$, output (units/hour):

L	1	2	3	4	5
Q	30	52	68	78	83

1. Compute the discrete MPL for each step.
2. Where do you see diminishing marginal returns?

Solution

$$MPL_{1 \rightarrow 2} = 52 - 30 = 22$$

$$MPL_{2 \rightarrow 3} = 68 - 52 = 16$$

$$MPL_{3 \rightarrow 4} = 78 - 68 = 10$$

$$MPL_{4 \rightarrow 5} = 83 - 78 = 5$$

Interpretation

With K fixed, the output increment per worker falls: congestion/bottleneck.

Exercise—MPK and capacity

With $L = 4$ fixed, output:

K	1	2	3	4
Q	78	110	130	140

1. Compute the discrete MPK .
2. Over which range does K “pay off the most”?

Solution

$$MPK_{1 \rightarrow 2} = 110 - 78 = 32$$

$$MPK_{2 \rightarrow 3} = 130 - 110 = 20$$

$$MPK_{3 \rightarrow 4} = 140 - 130 = 10$$

Interpretation

The first capacity expansion “unlocks” output. Then the constraint becomes labor/organization.

Cobb–Douglas function

Cobb–Douglas

$$Q = AL^{\alpha}K^{\beta}$$

$$MPL = \alpha AL^{\alpha-1}K^{\beta}, \quad MPK = \beta AL^{\alpha}K^{\beta-1}$$

Remember

With $\alpha < 1$ and K fixed, $L^{\alpha-1}$ decreases as L increases $\Rightarrow MPL$ tends to fall.

Exercise—Cobb–Douglas

Given:

$$Q = L^{0.6} K^{0.4}$$

1. Compute MPL and MPK .
2. Evaluate MPL at $(L, K) = (10, 5)$.
3. If L increases (with K fixed), what happens to MPL ? (1 line)

Solution

$$MPL = 0.6L^{-0.4}K^{0.4}, \quad MPK = 0.4L^{0.6}K^{-0.6}$$

At (10, 5):

$$MPL = 0.6 \cdot 10^{-0.4} \cdot 5^{0.4}$$

Interpretation

With K fixed, $L^{-0.4}$ decreases as L increases $\Rightarrow MPL$ tends to fall (diminishing marginal returns).

Exercise—Managerial decision

A firm has fixed K (given plant capacity) and observes that MPL is very low. It wants to increase output by 10% in the short run.

Question

Is it more plausible to do so by hiring alone, or by also investing in capacity? Argue in 3 lines.

Solution

If MPL is very low, 1 more worker adds little output: the constraint is capacity (K) or organization. Investing in K can remove the bottleneck and raise labor productivity (including MPL).

Remember

For significant increases in Q , a mix is often needed: **capacity + labor**, not labor alone.

Watch out!

In the short-run model K is fixed: only L can be adjusted. So the 10% increase can be achieved *only* by increasing L , but if MPL is very low it may require a very large increase in labor (and hence not be profitable/realistic).

Overall summary

- Technology: $Q = f(L, K)$
- Marginal products: MPL , MPK (output increments)
- Short run: fixed input $\Rightarrow$ typically diminishing marginal returns
- Graphs: TP rises but with a decreasing increment; MP vs AP relationship

Next lecture

Isoquants, MRTS and returns to scale: how to substitute labor and capital to produce the same Q .

Skills acquired

- ✓ Define technology and the production function
- ✓ Compute MPL and MPK (from a table or from a function)
- ✓ Recognize and explain diminishing marginal returns
- ✓ Connect concepts to decisions: capacity vs hiring

Optional homework exercises

1. With $Q = L^{0.5}K^{0.5}$ compute MPL and MPK . What happens to MPL if you double L with K fixed?
2. From the production table, compute AP and verify the MP vs AP rule.
3. Explain why in the short run “hiring too much” can reduce efficiency (coordination costs).